

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Term: Quiz 1 (SET 2)

Course Number: Math 4241

Course Title: Integral Calculus and Differential Equations

Full Marks: 15

Time: 30 Minutes

Answer all the questions. The symbols have their usual meanings. Marks of each question are written in the brackets.

1. a)	Define antiderivative. Use the five representative rectangles to find two approximations, (i) right-endpoint and (ii) left-endpoint of the area of the region lying between the graph of $f(x) = -x^2 + 5$ and the x-axis between $x = 0$ and $x = 2$.	(1+3+3)
b)	Define first and second fundamental theorems of integral calculus. Derive the reduction formula for $\cos^n x$.	(2+3)
	Define Gamma and Beta functions.	(2)
c)	Write down the order and degree of the following differential equations, also mention if the differential equation is linear or non-linear. (i) $(x+1)\frac{dy}{dx} - y = e^x(x+1)^2$ (ii) $(x^4 + 2y^3)\frac{d^2y}{dx^2} + (3x^2 - x)\frac{dy}{dx} + y = 0$ (iii) $\left(\frac{d^3y}{dx^3}\right)^2 + x\frac{d^2y}{dx^2} = x^2$ (iv) $\frac{d^3y}{dx^3} + x\frac{d^2y}{dx^2} + x\frac{dy}{dx} + 2y = e^x$	(1+1+1+1)
d)	Form the differential equation of the given solution $y = A \cos x + B \sin x$	(2)

Name:**Student ID:**

(Start writing from the line below)